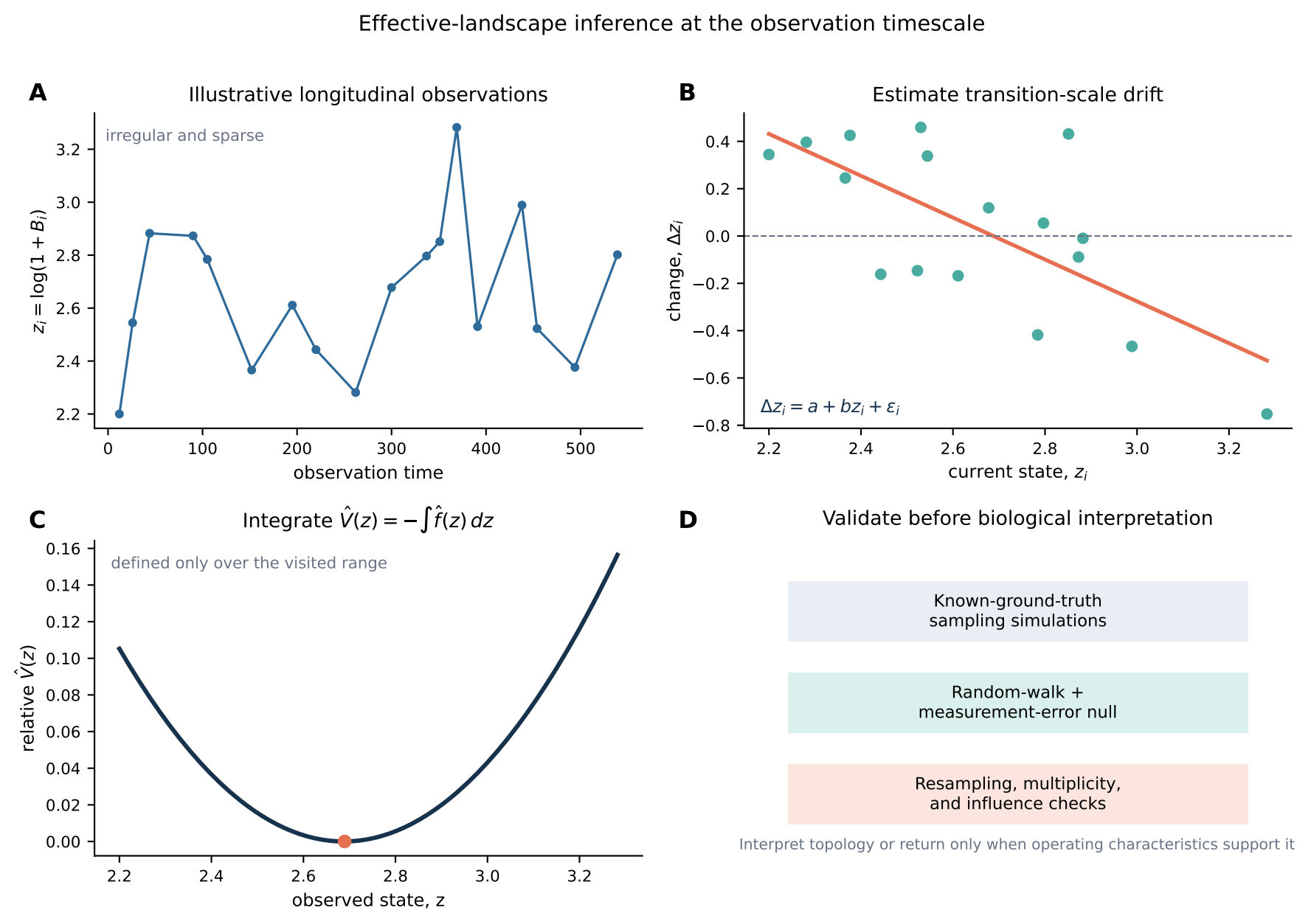
Effective-landscape inference at the observation timescale
A
Illustrative longitudinal observations
irregular and sparse
$z_i = \log(1 + B_i)$
observation time
B
Estimate transition-scale drift
change, Δz_i
current state, z_i
$\Delta z_i = a + bz_i + \varepsilon_i$
C
Integrate $\hat{V}(z) = -\int \hat{f}(z)\,dz$
defined only over the visited range
relative $\hat{V}(z)$
observed state, z
D
Validate before biological interpretation
Known-ground-truth sampling simulations
Random-walk + measurement-error null
Resampling, multiplicity, and influence checks
Interpret topology or return only when operating characteristics support it